Module 30 - Dense vs Sparse Arrays

1. What is Array?

An array is a data structure used to store multiple values in a single variable. Each value is stored at a specific index, starting from index 0.

Example Code:

```
// Array
let flavors = ["Vanilla", "Butterscortch", "Lavendar", "Chocolate"];
// Accesses array elements using index numbers.
console.log(flavors);
console.log(flavors[0]);
console.log(flavors[1]);
console.log(flavors[2]);
console.log(flavors[3]);
// Returns the last element.
console.log(flavors[flavors.length - 1]);
```

2. Rules & Features of Arrays

- An array is a data structure used to store multiple values in a single variable.
- An array can store different data types (numbers, strings, booleans, objects, etc.).
- An array can store homogeneous (same data type) or heterogeneous (different data types) values.
- Array elements are accessed using their index, and indexing starts from 0.
- Arrays are dynamic (flexible) in size, so they can grow or shrink when elements are added or removed.

3. Types of Array Creation

- Array Literal
- Array Constructor

4. What is Array Literal?

Array Literal is the simplest and most commonly used way to create an array using square brackets [].

Example Code:

```
// Array Literal
let sample = [1, "two", true, null, undefined, { id: 1 }];
console.log(sample);
```

5. What is Array Constructor?

Array Constructor is a way to create an array using the new Array() constructor.

Example Code:

```
// Array Constructor
let newArray = new Array();
// Adds values to the array using index numbers.
newArray[0] = "First";
newArray[1] = "Second";
newArray[2] = "Third";
newArray[3] = "Fourth";
console.log(newArray);
// Returns the total number of elements in the array.
console.log(newArray.length);
```

6. Dense vs Sparse Arrays

- **Dense Array**: An array in which every index contains a value, with no empty spaces.
- **Sparse Array**: An array in which one or more indexes are empty (missing values).

Example Code:

```
// Dense Array
let denseArray = [1, 2, 3, 4, 5]; // Uses contiguous memory locations.
// Example Base Addresses: 1004, 1008, 1012, 1016, 1020
// Memory Address = Base Address + (Index * Size)
//                = 1004 + (0 * 4) = 1004
console.log(denseArray);


// Sparse Array
let sparseArray = [10, 20, , 40, , 60]; // Uses a Hash Table or Hash Map internally.
console.log(sparseArray);
```